

Q1. Solve the system

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array} \right)$$

$$R2 \leftarrow R2 + 1.5 R1, \quad R3 \leftarrow R3 + R1$$

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 0.5 & 0.5 & 1 \\ 0 & 2 & 1 & 5 \end{array} \right)$$

$$R3 \leftarrow R3 - 4 R2$$

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & 0.5 & 0.5 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right)$$

$$\text{back-sub: } z = -1, \quad 0.5y - 0.5 = 1 \Rightarrow y = 3, \quad 2x + 3 + 1 = 8 \Rightarrow x = 2$$

$$x = \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$$

Q2. Rank and nullity

rref has 2 pivots, so rank $A = 2$

$$\text{nullity} = 3 - 2 = 1$$

$$\text{null space: } (-1, 2, 1)$$

Q3. Subspace proof

$$W = \{ (x, y, z) : x + 2y - z = 0 \}$$

Need: closure under + and under scalar multiplication.

Take $(1, 0, 1)$ and $(0, 1, 2)$, both in W .

Sum is $(1, 1, 3)$: $1 + 2 - 3 = 0$, so in W .

$3 \cdot (1, 0, 1) = (3, 0, 3)$: $3 + 0 - 3 = 0$, in W .

For arbitrary u, v in W and c in $\mathbb{R}$ the same identity holds by linearity:

$$(u_1 + v_1) + 2(u_2 + v_2) - (u_3 + v_3)$$

$$= (u_1 + 2u_2 - u_3) + (v_1 + 2v_2 - v_3) = 0 + 0 = 0$$

$$c u_1 + 2 c u_2 - c u_3 = c (u_1 + 2u_2 - u_3) = c \cdot 0 = 0$$

so W is a subspace of $\mathbb{R}^3$.

Q4. Inverse

$$[A|I] =$$

$$\left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$R1 \leftarrow R1 - R2$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$R2 \leftarrow R2 - R3$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right)$$

$$A^{-1} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\text{Check: } A A^{-1} =$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I \checkmark$$